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C - Buy an Integer

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Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 300 points

Problem Statement

Takahashi has come to an integer shop to buy an integer.

The shop sells the integers from 1 through 10^9 . The integer N is sold for $A \times N + B \times d(N)$ yen (the currency of Japan), where $d(N)$ is the number of digits in the decimal notation of N .

Find the largest integer that Takahashi can buy when he has X yen. If no integer can be bought, print 0.

Constraints

- All values in input are integers.
- $1 \leq A \leq 10^9$
- $1 \leq B \leq 10^9$
- $1 \leq X \leq 10^{18}$

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Input

Input is given from Standard Input in the following format:

```
A B X
```

Output

Print the greatest integer that Takahashi can buy. If no integer can be bought, print 0.

Sample Input 1

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```
10 7 100
```

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Sample Output 1

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```
9
```

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The integer 9 is sold for $10 \times 9 + 7 \times 1 = 97$ yen, and this is the greatest integer that can be bought. Some of the other integers are sold for the following prices:

- $10 : 10 \times 10 + 7 \times 2 = 114$ yen
- $100 : 10 \times 100 + 7 \times 3 = 1021$ yen
- $12345 : 10 \times 12345 + 7 \times 5 = 123485$ yen

Sample Input 2

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```
2 1 1000000000000
```

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Sample Output 2

[Copy](#)

```
10000000000
```

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He can buy the largest integer that is sold. Note that input may not fit into a 32-bit integer type.

Sample Input 3

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```
10000000000 10000000000 100
```

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Sample Output 3

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```
0
```

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Sample Input 4

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```
1234 56789 314159265
```

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Sample Output 4

Copy

```
254309
```

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Language

C++23 (Clang 21.1.0)

Source Code

Open FileCustomizeToggle EditorAuto Height

```
1 |
```

* at most 512 KiB

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#telegram)

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